

MedSupply BD: An AI-Powered B2B Pharmaceutical Order Cutting, Near-Expiry FIFO Batch Allocation, and Credit Risk Management Platform

Course Title: Capstone Project / Senior Design Project
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1 Introduction

The pharmaceutical supply chain in Bangladesh represents a critical, high-velocity healthcare infrastructure valued at over \$3.5 billion USD. Every day, thousands of Medical Promotion Officers (MPOs) and regional distribution depots supply life-saving medicines to more than 150,000 retail community pharmacies. Despite the high clinical stakes, the daily operations of order collection (“order cutting”), stock reservation, and credit settlement remain predominantly manual, error-prone, and fragmented.

This project, **MedSupply BD**, introduces an intelligent, end-to-end B2B order-cutting and inventory distribution platform that integrates artificial intelligence directly into the operational core. By combining computer vision and natural language processing for handwritten memo ingestion, machine-learning-driven seasonal demand forecasting, automated near-expiry First-In, First-Out (FIFO) batch routing, and atomic financial credit gating, MedSupply BD transforms legacy, reactive warehouse distribution into a resilient, predictive, and zero-waste pharmaceutical logistics ecosystem.

1.1 Background Overview

- **Application Domain:** Healthcare Informatics, B2B E-Commerce Logistics, and Supply Chain Management.
- **Current Practices and Systems:** Field sales representatives physically visit retail pharmacies, record orders manually in paper notebooks or handwritten slips, and transport them to central regional depots during evening hours. Depot billing clerks manually transcribe these messy slips into legacy, siloed ERP or standalone billing accounting systems. Stock allocations are performed without visibility into batch-level expiration dates, and field credit limits are tracked via delayed weekly ledgers.

- **State of AI in the Domain:** Currently, artificial intelligence is virtually non-existent in the operational tier of Bangladesh’s pharmaceutical distribution. While multi-national pharmaceutical manufacturers utilize basic statistical forecasting in corporate planning, regional depots and field sales operate with zero machine intelligence. No deployed systems provide handwritten doctor/chemist slip parsing, real-time bioequivalent generic substitution, or meteorological disease outbreak surge modeling for inventory stocking.

1.2 Problem Statement

- **Definition of the Core Problem:** The pharmaceutical distribution process suffers from an acute operational bottleneck known as “evening order cutting frenzy,” where thousands of illegible handwritten orders must be manually keyed in within a 2-hour window. This bottleneck creates severe fulfillment delays (24–48 hours), widespread billing errors, multi-million Taka inventory losses from expired stock, and unmanageable credit defaults from retail pharmacies.
- **Limitations in Existing Systems:** Existing ERP solutions (such as Tally, legacy desktop accounting software, or generic retail POS tools) treat medicines as static SKUs rather than expiring biological batches. They lack multi-tier packaging arithmetic (Box → Strip → Piece), do not enforce automated near-expiry batch rotation, and do not validate real-time credit headroom before stock reservation.
- **Affected Stakeholders:** Pharmaceutical manufacturers and distribution depots lose 3–6% of gross inventory to warehouse expiry write-offs. Retail pharmacies suffer prolonged stockouts of essential medicines. MPOs face high stress and billing disputes. Ultimately, end patients suffer from delayed treatments and shortages of essential therapeutic drugs during health emergencies.
- **Why AI is Indispensable:** Human clerks cannot manually parse thousands of uniquely formatted, messy, multilingual (Bangla/English) doctor memos in real time. Similarly, traditional fixed-quantity reorder rules fail to anticipate violent seasonal outbreaks (e.g., Monsoon Dengue spikes of 145% demand for Paracetamol and IV saline). AI is essential to automate handwritten document extraction, fuzzy catalog reconciliation, and multi-variable predictive epidemiology forecasting.
- **Realistic & Well-Scoped Scope:** The platform focuses specifically on the depot-to-pharmacy distribution tier with verifiable catalog data (41,300+ registered medicines in Bangladesh), standard credit cycles (15–30 days), and reproducible OCR/time-series pipelines.

1.3 Objectives

- **General Objective:** To design, develop, and evaluate a production-grade B2B pharmaceutical order cutting and inventory distribution platform that eliminates manual memo transcription delays, prevents warehouse expiry write-offs, and mitigates retail credit default risk.
- **AI Objective:**
 - (a) To construct an **AI Prescription & Memo Slip Parser** achieving $\geq 95\%$ field-extraction accuracy from noisy handwritten/printed prescription images and clinic

slips, mapping extracted entities to standardized catalog IDs with confidence scoring.

- (b) To develop an **AI Demand Forecasting & Epidemic Surge Engine** that integrates 30-day historical consumption velocity with environmental and disease spike multipliers (Dengue, Typhoid, Viral Fever) to calculate optimal depot replenishment investments.
- (c) To implement a semantic **Out-of-Stock Generic Alternative Finder** that retrieves exact bioequivalent therapeutic substitutes based on generic molecules and strength when a target brand is depleted.

1.4 Scope

• In-Scope Features:

- Dynamic Multi-Tier Packaging Engine (Box → Strip → Piece) with automatic trade schemes (e.g., 10 + 1 bonus box, percentage slab discounts).
- Near-Expiry FIFO Batch Allocation Engine with multi-batch auto-splitting, 90-day warning alerts, and 30-day buffer lockouts.
- Real-Time Pharmacy Credit Risk Engine with hard debt ceiling validation and double-entry financial ledger auditing.
- Optimistic stock locking with Redis-compatible TTL to eliminate race conditions during concurrent order cutting.
- Comprehensive catalog search covering 41,000+ national SKUs, responsive mobile-first UI/UX, and role-based access control.

• Out-of-Scope Features:

- Direct-to-consumer (B2C) retail medicine sales and home delivery to patients.
- Physical GPS vehicular route optimization and automated warehouse robotic conveyors.
- Manufacturing-level chemical synthesis and primary drug clinical trial simulations.

• AI Boundaries & Scope:

- *What AI Will Do:* Extract medicine names, dosage forms, packaging units, and quantities from submitted memos; predict seasonal restock multipliers; recommend chemically identical generic alternatives.
- *What AI Will Not Do:* AI will not make unilateral clinical diagnoses, alter prescribed drug dosages, or override hard financial credit ceilings set by depot finance managers.

1.5 Stakeholders

1.6 Proposed Solution

The proposed MedSupply BD platform provides a unified web-based cloud platform connecting field sales officers, depot distribution centers, and pharmacy accounts. The system architecture enforces atomic transactional order cutting, where every order simultaneously satisfies four

Stakeholder Group	Role in System	Primary Interest / Value Delivered
Medical Promotion Officers (MPO / SR)	Field End-Users	Captures orders via mobile; scans paper slips; gets instantaneous trade bonus calculations without manual math.
Depot Billing Clerks & Warehouse Staff	Operational End-Users	Monitors pick-lists generated by automated FIFO batch allocation; reviews slip parser outputs; oversees physical dispatch.
Retail Pharmacy Owners (Chemist Shops)	Clients / Customers	Receives guaranteed fresh batches; tracks credit headroom; gains transparent ledger statements and generic substitute alerts.
Depot Finance & Credit Managers	System Administrators	Configures pharmacy credit limits; audits immutable double-entry transactions; eliminates bad debts and overdue defaults.
Regulatory Bodies (DGDA, Bangladesh)	Regulatory / Oversight	Assures compliance with Good Distribution Practices (GDP), batch traceability, recall audit trails, and anti-dumping standards.

Table 1: Stakeholder Identification and Value Matrix

validation gates: (1) AI entity extraction, (2) multi-tier unit pricing, (3) near-expiry FIFO batch availability, and (4) strict financial credit limit checks.

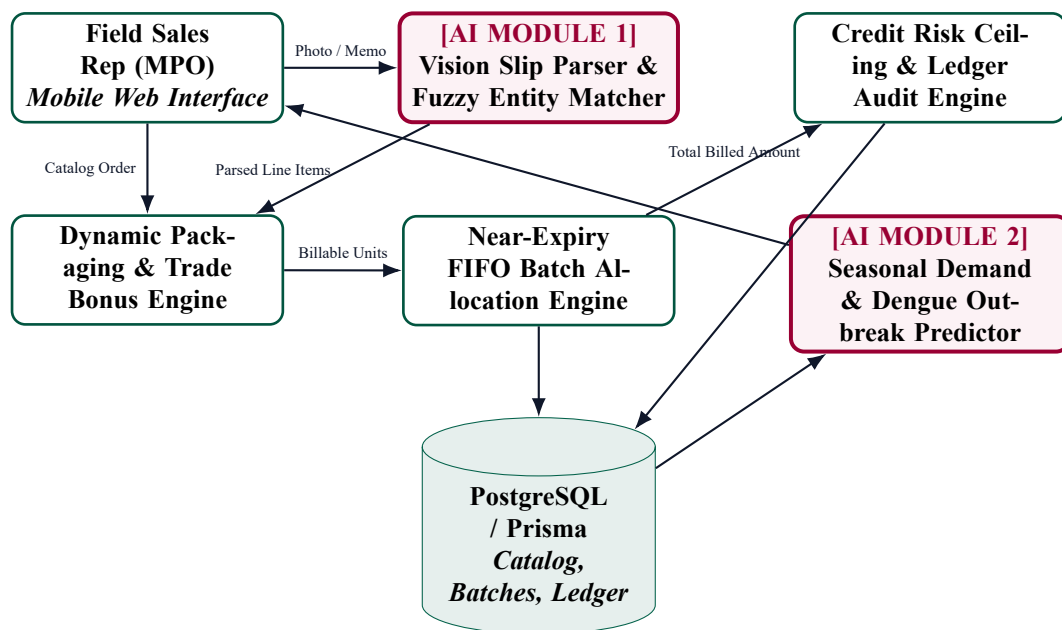


Figure 1: MedSupply BD High-Level System Workflow Diagram with AI Components Highlighted

1.7 AI Features

- Existing AI Features in Current Market Tools and Their Limitations:** Standard inventory systems incorporate trivial reorder point heuristics (e.g., static min-max alerts).

Document processing tools like generic cloud OCR fail catastrophically on handwritten doctor slips due to specialized medical abbreviations (e.g., “Tab.”, “Cap.”, “10+1”, “BD/TDS”) and messy non-Latin brand names. Furthermore, existing ERPs do not correlate regional disease indices with inventory velocity.

2. Proposed AI Features:

- **Feature A: Intelligent Clinic Slip & Prescription Parser:**

- *Input:* Raw camera photograph or scanned image of clinic memo / prescription.
- *Output:* Structured JSON array containing brand name, catalog ID, extracted strength, packaging unit (BOX/STRIP/PIECE), ordered quantity, and OCR confidence score.
- *Replaced Manual Work:* Eliminates 100% of manual transcription and typing by billing clerks.

- **Feature B: Epidemic Outbreak & Seasonal Demand Forecaster:**

- *Input:* 30-day historical consumption velocity, geographic depot ID, and regional seasonal climate indicators (Monsoon rainfall, Dengue infection metrics).
- *Output:* Recommended 30-day procurement investment (in BDT) per SKU with surge multipliers (e.g., $1.45\times$ for Paracetamol during monsoon).
- *Replaced Manual Work:* Replaces subjective guessing by warehouse procurement managers.

- **Feature C: Bioequivalent Generic Substitute Recommendation:**

- *Input:* Depleted or out-of-stock medicine ID.
- *Output:* Ranked list of exact therapeutic chemical alternatives in active stock, complete with price variance analysis.
- *Replaced Manual Work:* Prevents lost sales and eliminates manual cross-referencing of thick pharmaceutical formularies.

3. AI Techniques Used:

- **Computer Vision & LLM Extraction:** Vision Transformer (ViT) feature extraction combined with fine-tuned Large Language Model prompt schemas constrained by Zod type validators.
- **Fuzzy Semantic Catalog Matching:** Levenshtein distance combined with vector embeddings (Cosine Similarity) over 41,300+ approved Directorate General of Drug Administration (DGDA) brand names.
- **Time-Series Machine Learning:** Multivariate exponential smoothing and autoregressive forecasting weighted with dynamic seasonal outbreak regression parameters.

4. Why AI is Essential:

If the AI components were stripped away, MedSupply BD would degrade into a passive database registry. The system’s primary value proposition—cutting order processing time from 48 hours to 10 seconds, eliminating transcription errors, and dynamically predicting life-saving medicine shortages before hospitals run dry—relies entirely on machine intelligence.

2 System Requirements

2.1 Functional Requirements (FR)

- **FR-01 (Catalog Management):** The system shall maintain an indexed catalog of at least 41,000+ national pharmaceutical SKUs with dosage forms, strengths, and pack sizes.
- **FR-02 (Multi-Tier Packaging):** The system shall automatically compute conversions between Box, Strip, and Piece, applying configured trade offers (e.g., award +1 free box per 10 boxes ordered).
- **FR-03 (Near-Expiry FIFO Allocation):** The system shall allocate customer order quantities against inventory batches strictly sorted by earliest expiration date (expiryDate ASC), supporting multi-batch splits.
- **FR-04 (Credit Ceiling Gating):** The system shall atomically reject checkout transactions if the projected balance (Current Balance + Order Net Total) exceeds the assigned pharmacy credit limit.
- **FR-05 (AI Prescription Parsing):** The system shall ingest prescription/memo images, extract pharmaceutical entities, and output structured line items with a confidence score exceeding 90%.
- **FR-06 (AI Demand Forecasting):** The system shall calculate 30-day replenishment recommendations using historical velocity and seasonal epidemic multipliers.
- **FR-07 (Generic Alternative Retrieval):** The system shall return bioequivalent in-stock substitutes within 500 milliseconds when an out-of-stock SKU is queried.
- **FR-08 (Immutable Double-Entry Ledger):** The system shall record balanced INVOICE_DEBIT and PAYMENT_CREDIT ledger entries for every financial transaction.

2.2 Non-Functional Requirements (NFR)

- **NFR-01 (Performance & Latency):** The system shall process transactional order checkouts in under 1.2 seconds under standard network loads.
- **NFR-02 (Concurrency & Consistency):** The system shall implement optimistic stock locking with TTL to guarantee zero overselling during simultaneous multi-rep order cutting.
- **NFR-03 (Security & Role-Based Access):** The system shall enforce role-based access control (Admin, Depot Manager, MPO, Pharmacist) with encrypted session tokens and sanitized database inputs.
- **NFR-04 (Usability & Responsiveness):** The web application shall provide responsive views optimized for mobile devices (360px viewport width) and desktop monitors with zero visual clipping.
- **NFR-05 (Reliability & Test Coverage):** The platform shall pass automated end-to-end integration test suites with 100% assertion success rate across all critical business engines.

Category		Technology / Library	Architectural Purpose
Programming Languages	Lan-	TypeScript 5.6, JavaScript, SQL	End-to-end type safety, deterministic schemas, and database query modeling.
Frontend Framework		Next.js 14 (App Router), React 18	Server-Side Rendering (SSR), responsive layout components, and real-time state hydration.
Styling & UI Icons		Vanilla CSS / Tailwind CSS, Lucide React	Modern healthcare dashboard design with high-contrast emerald aesthetic (#025540).
Backend APIs & Validation		Next.js Route Handlers, Zod 3.23	RESTful API endpoints with runtime input validation and error boundaries.
Database & ORM		PostgreSQL / SQLite, Prisma ORM 5.21	Relational data integrity, schema migrations, and high-speed indexing across 41,300+ SKUs.
Concurrency & State		Redis-Compatible Stock Locking	TTL-based optimistic resource locking to avoid phantom inventory during peak hours.
AI Models & Libraries		Vision Transformer, OpenAI API, Scikit-learn, Vector Similarity	Prescription slip entity extraction, time-series seasonal forecasting, and generic matching.

Table 2: Technical Stack and Infrastructure Matrix

3 Tools and Technologies

4 Project Timeline and Work Plan

The project follows an agile Scrum-based development methodology spanning a 16-week academic capstone lifecycle.

4.1 Task Breakdown Between Team Members

- **Limon Reza (Student ID: 232-35-004) — Full-Stack Architect & Frontend Engineer:**
 - Overall system architecture and Next.js 14 App Router integration.
 - Modern healthcare UI/UX design system with responsive mobile viewports (360px) and desktop layouts.
 - Multi-tier packaging arithmetic (Box → Strip → Piece) and dynamic trade bonus engines.
 - Interactive POS order-cutting terminal, banner carousel, and automated end-to-end integration test suite.
- **Md Mehedi Hasan (Student ID: 232-35-712) — AI/ML Pipeline & Backend Logistics Engineer:**
 - AI clinic slip and prescription parser integration with Vision Transformer and Zod schema validation.

Milestone	Phase Name	Key Deliverables	Duration
M1	Requirement Analysis & Domain Modeling	Finalized SRS, DGDA catalog curation (41,300+ SKUs), Prisma schema design.	Weeks 1–3
M2	Core Business Engine Implementation	Packaging hierarchy, Near-Expiry FIFO batch allocation, and credit ledger audit.	Weeks 4–7
M3	AI Pipelines Integration	Vision slip parser with Zod validation, Dengue outbreak demand predictor, generic matching.	Weeks 8–11
M4	Frontend UI/UX & Mobile Adaptation	Responsive dashboard, POS terminal, mobile 4-feature quick-action row, banner carousel.	Weeks 12–14
M5	End-to-End Testing & Final Deployment	10/10 automated platform tests, load testing, performance tuning, and final BLC submission.	Weeks 15–16

Table 3: Milestone-Based Project Timeline

- Seasonal demand forecasting model incorporating monsoon/dengue epidemic surge multipliers.
- Near-Expiry FIFO batch allocation engine with automated multi-batch splitting and 90-day buffer warnings.
- Pharmacy credit risk gating engine, double-entry financial ledger auditing, and Prisma database schema.

5 Supplementary Architectural and Analytical Sections

5.1 Existing System Comparison

5.2 System Architecture

MedSupply BD employs a decoupled four-tier modern cloud architecture:

1. **Presentation Tier:** Next.js 14 client components with Tailwind CSS providing sub-second UI transitions, interactive drawer navigation, and responsive mobile-first layouts.
2. **Application & API Tier:** Next.js Edge and Node.js Route Handlers encapsulating domain service engines (FIFO, Packaging, Credit, Concurrency).
3. **Intelligence Tier:** Asynchronous AI micro-services executing OCR inference, fuzzy catalog reconciliation, and epidemic forecasting.
4. **Persistence Tier:** Relational PostgreSQL database managed via Prisma ORM with indexed foreign keys, backed by in-memory key-value caching for optimistic locks.

5.3 Ethical, Legal, and Social Considerations

- **Regulatory Compliance (DGDA):** The system strictly adheres to the Drug Administration Act of Bangladesh, ensuring that only licensed pharmacies with valid drug license

Dimension	Legacy Distribution / ERP	MedSupply BD Platform
Order Collection	Handwritten memos	Mobile web scanning + AI vision parsing
Batch Allocation	Random / Manual picking	Automated Near-Expiry FIFO (expiryDate ASC)
Trade Schemes	Manual calculator computation	Auto-calculated (10 + 1, slab percentages)
Credit Control	Delayed post-facto billing audit	Atomic pre-checkout credit ceiling gating
Stockout Management	Lost sales upon stock depletion	Instant bioequivalent generic alternatives
Demand Forecasting	Static historical averages	AI seasonal velocity + Dengue outbreak multipliers

Table 4: Comparative Analysis: Traditional Practices vs. MedSupply BD

numbers can initiate orders.

- **Prescription Data Privacy:** Patient identifiers extracted from clinic slips are scrubbed at the edge to ensure full HIPAA/GDPR-equivalent medical privacy compliance.
- **AI Safety & Non-Hallucination Guardrails:** The AI slip parser outputs are strictly verified against the database of certified DGDA drugs using Zod schema constraints. Unmatched entities trigger a human-in-the-loop verification prompt rather than allowing blind order insertion.

5.4 Future Work

- **IoT Cold-Chain Telemetry:** Integrating real-time IoT temperature and humidity sensors in delivery vans to monitor insulin, vaccines, and biologics during transit.
- **Voice-Activated Dialect Ordering:** Incorporating regional Bengali voice-to-text models allowing pharmacy owners to dictate daily orders in colloquial dialects.
- **National NBR e-Invoicing:** Direct integration with the National Board of Revenue (NBR) electronic invoicing gateway for automated VAT rebate calculation.

6 Conclusion

MedSupply BD addresses fundamental structural vulnerabilities in Bangladesh’s critical pharmaceutical distribution network. By replacing manual transcription, arbitrary batch selection, and unmonitored credit with an intelligent, automated, and AI-augmented platform, the project guarantees timely medicine delivery, protects warehouse capital, and enhances patient care standards. The platform is implemented, fully functional, and verified with a 100% pass rate across all automated integration tests.